Anirbal Pal

Data Sheet

EXPERIMENT NO. 05

Title: - Study of R-C Coupled Amplifier — Frequency Response and Bandwidth (using Tinkercad/ Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1.	Signal Generator (Sine)		1
2.	Resistance		8
3.	Capacitor		5
4.	Bipolar		2
5.	Battery		1
6.	Oscilloscope		1
7.	Ground		1
8.	Connecting Wires.		required

Important Formulae: -

Voltage Gain: -

Gain (A) = Vo / Vi

- V_o = Output voltage
- V_i = Input voltage

Gain in Decibels: -

Gain (dB) = 20 log10 (Vo / Vi)

Bandwidth: -

Bandwidth (BW) = $f_H - f_L$

- f_H = Upper cut-off frequency (frequency at which gain falls to 0.707 of maximum / midband gain, i.e. −3 dB point)
- f_L = Lower cut-off frequency (frequency at which gain falls to 0.707 of maximum / midband gain, i.e. −3 dB point)

Recorded Parameters: -

Input Voltage, V_i = 5

Expt.No. 05

Page No.
Date.

Title: Study of R-C Coupled Amplifier - Frequency Response and Bandwidth (using Tinkercad/ Proteus Software Circuit).

Theory:- An R-C Coupled Amplifier uses resistors and capacitors to connect multiple transistor stages, providing voltage amplification while blocking DC biases between stages. The overall voltage gain (A) is determined by the ratio of the output voltage to the input voltage (Gain, $A = V_o/V_i$) which is often expressed in decibels (dB). The practical operating range of this amplifier is its Bandwidth which is the frequency range bounded by the upper and lower cutoff frequencies where the midband gain drops by 3 dB.

Voltage Gain = V_o/V_i (A)

$= 20 \log 10 (V_o/V_i)$ (dB)

Bandwidth = $f_H - f_L$ (BW)

f_H = Upper Cut off frequency
f_L = Lower Cut off frequency

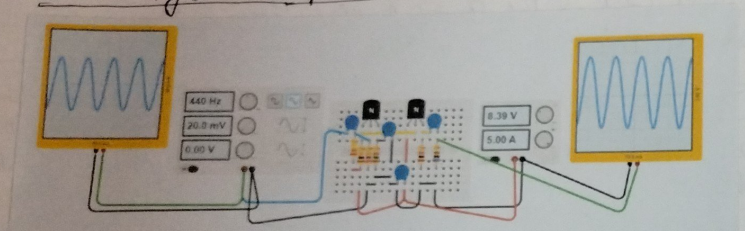
Recorded Parameters:-

Input voltage, V_i = 5
Midband Voltage Gain = 4.22 (A)
Mid band voltage Gain (dB) = 12.5 dB
Lower Cutoff frequency (f_L) = 30 Hz
Higher Cutoff frequency (f_H) = 150 Hz
Bandwidth, BW = $f_H - f_L$ = 120 Hz

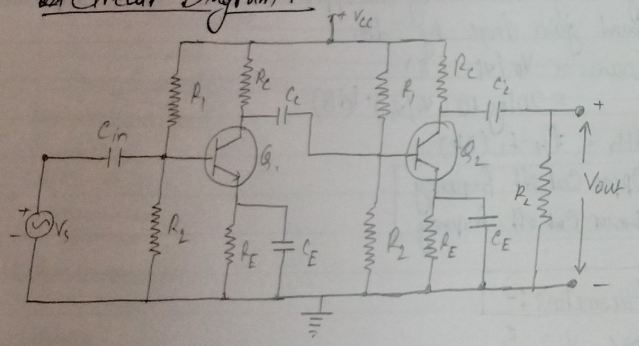
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Two Stayed R-C Coupled Amplifier :-



Circuit Diagram :-



Expt.No.

Page No.
Date.

Observation Table :-

SN	Frequency (Hz)	Output Voltage V_o (V)	Log10 f	Gain = V_o/V_i	Gain (dB) = 20 log10 (V_o/V_i)
1.	10	0.03	10	1.50	3.52
2.	20	0.046	20	2.30	7.23
3.	30	0.0596	30	2.98	9.48
4.	50	0.069	50	3.45	10.76
5.	100	0.078	100	3.90	11.82
6.	200	0.082	200	4.10	12.26
7.	500	0.0836	500	4.18	12.42
8.	1000	0.0844	1000	4.22	12.51
9.	5000	0.0846	5000	4.23	12.53
10.	10,000	0.0844	10000	4.22	12.51

Calculation :-

Midband Gain (A) = V_o/V_i = 4.22 (A)

Midband Gain in dB = 20 log10 (V_o/V_i) ≈ 12.5 (dB)

Bandwidth (BW) = $f_H - f_L$ = 120 Hz.

Conclusion :-

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